

Text Classification in NLP

CMPE346 · Group 2

Atakan Gül [Student 2] [Student 3]

June 5, 2026

Group task

Text Classification

Metric

Accuracy & Weighted F1

Student	Task
Atakan Gül	Media Bias Detection
[Student 2]	[Topic]
[Student 3]	[Topic]

BiLSTM

Bidirectional LSTM trained from scratch. Captures sequential context in both directions.

BERT `bert-base-uncased`

Transformer encoder pretrained via masked language modeling. Fine-tuned end-to-end.

RoBERTa `roberta-base`

Optimized BERT: removes next sentence prediction, uses dynamic masking and more data.

Task

Classify a news sentence as *biased* or *not biased*.

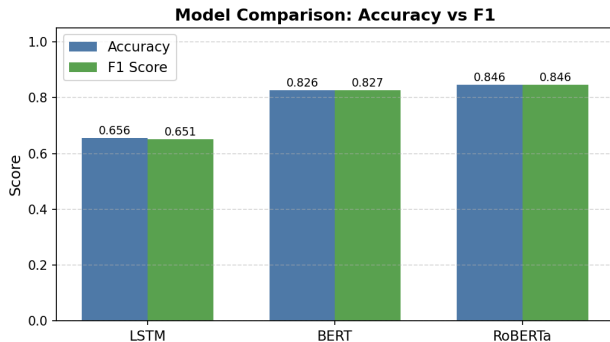
Dataset: BABE

- 4,121 expert-annotated sentences
- Diverse news outlets
- 3,121 train / 1,000 test

Atakan — Results

Model	Acc.	W-F1
BiLSTM	0.6560	0.6511
BERT	0.8260	0.8266
RoBERTa	0.8460	0.8464

- RoBERTa best: +19pp accuracy over BiLSTM
- BERT close behind
- BiLSTM struggles on subtle bias



Atakan — Prediction Examples

Sentence	True	LSTM	BERT	RoBERTa
<i>“Nearly 78% of Americans report concern about the climate...”</i>	NB	NB ✓	NB ✓	NB ✓
<i>“Companies like Nike and Google made June-teenth a paid holiday.”</i>	NB	B ×	NB ✓	NB ✓
<i>“None of Biden’s Democrat counterparts challenged his claim of economic selflessness.”</i>	B	B ✓	NB ×	B ✓

NB = Not Biased B = Biased ✓ = correct × = incorrect

- Media bias is expressed through framing and tone — not facts
- Pretraining is critical: transformers capture patterns BiLSTM cannot
- RoBERTa outperforms BERT by 2pp — modest but consistent with its optimized pretraining

Task	Dataset: [Name]
[Description]	Stats
	Train / test split

[Student 2] — Results

Model	Acc.	F1
BiLSTM	—	—
BERT	—	—
RoBERTa	—	—

[Figure]

ling 1

ling 2

ding 1

ding 2

ding 3

Task	Dataset: [Name]
[Description]	Stats
	Train / test split

[Student 3] — Results

Model	Acc.	F1
BiLSTM	—	—
BERT	—	—
RoBERTa	—	—

[Figure]

ling 1

ling 2

ding 1

ding 2

ding 3

All Results

Task	Model	Acc.	W-F1
Media Bias	BiLSTM	0.6560	0.6511
	BERT	0.8260	0.8266
	RoBERTa	0.8460	0.8464
[Student 2]	BiLSTM	—	—
	BERT	—	—
	RoBERTa	—	—
[Student 3]	BiLSTM	—	—
	BERT	—	—
	RoBERTa	—	—

- RoBERTa consistently outperforms across all three tasks
- BERT and RoBERTa adapt well regardless of classification domain
- Pretraining is decisive — BiLSTM cannot compete on small datasets